

Worksheet 2B: Linear Thinking — Data & Curves — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Instructor Copy — Not for Distribution

Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. (a) Dataset A (b) Dataset B (c) Dataset C (d) ≈ 5 points per hour studied
2. (a) Dataset P (b) Positive; higher temp $\rightarrow$ more sales (c) ≈ 13 euros per degree (d) $\approx 500\text{€}$
3. (a) Dataset Y (b) $\approx 0.175^\circ\text{C}$ per decade (c) $\approx 1.4^\circ\text{C}$ (d) Natural variability, complex systems
4. (a) Dataset M (b) Dataset N (c) Accuracy can't exceed 1.0 (d) Between epochs 10-20

Part B

5. (a) Negative (b) Positive (c) Negative (d) Zero (e) Negative (f) Positive
6. (a) -2 (b) 0.5 (c) 0 (d) 2 (e) -0.5

Part C

7. (c) ≈ 1 (d) ≈ 1 (e) $y \approx x + 1$
8. (b) Predicted: 4, Actual: 5, Residual: +1 (c) Predicted: 5, Actual: 4, Residual: -1 (d) Yes, small balanced residuals
9. (a) Student B (b) Student B (c) Sum of squared residuals
10. Squaring makes all residuals positive; otherwise +5 and -5 would cancel to 0

Part D

11. (a) Flat (b) Positive (c) Negative (d) At $x = 0$
12. (a) Zero (b) Negative (c) Positive
13. (a) No (b) No (c) Steeper (d) As $x \rightarrow -\infty$ (leftmost part)
14. (a) Negative everywhere (b) At far left (small x) (c) Zero

15. (a) Up (positive) (b) Zero (c) Negative (d) Zero (e) Most positive: $x = 0, 2\pi$; Most negative: $x = \pi$
16. (a) Zero (b) No, always increasing or flat (c) Changing
17. (Sketching exercise - answers will vary)
18. (a) Forward (b) Slowing down (c) At the beginning (d) Car is stopped
19. (a) Positive (b) Zero (c) Negative (d) Velocity (speed and direction)
20. (Sketching exercise - examples: (a) $y = x^3$ (b) $y = e^x$ (c) $y = \sin(x)$)

Part E

21. (a) 13 (b) +1 (c) 1
22. (a) Strong positive (b) Strong negative (c) Very weak (d) Almost none
23. (a) Both caused by summer/heat (b) Larger fires need more firefighters (c) Both caused by age
24. (a) Each additional sq ft adds €200 to price (b) Base price/land value (c) Represents land + fixed costs (d) €450,000

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